

Trustworthiness Evaluation Framework

Model Configurations

Centralized, local-only, FedAvg, FedProx, DP, SecAgg simulation

Multi-Objective Evidence

Six trustworthiness dimensions

Predictive Utility

AUROC, AUPRC, F1
balanced accuracy, Brier score

Privacy

Membership-inference AUC
attack advantage

Explainability

SHAP attribution
PEDI drift

Fairness

City, area, and
service-category gaps

Calibration

ECE and
reliability curve

Efficiency

Runtime, memory,
communication cost

Component Normalization

Convert evidence to comparable 0-1 scores

TAI-Score

Weighted trustworthiness ranking

Transparency Record

Evidence, caveats, deployment risks